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1. Introduction

The project “**KrishiDisha**” is a database-driven agricultural learning, decision support, nutrition, and crop intelligence platform designed for agricultural students, researchers, and beginner farmers. The system provides an interactive web interface where users can explore detailed crop information, disease identification, profit calculation, consultation with experts, and nutrition-based food learning.

A major extension of this system is the **Crop Encyclopedia Module**, where users can view complete agricultural information about any crop with images. When a user clicks on a crop image, the system displays detailed information including scientific name, origin, availability in Bangladesh, cultivation history, crop varieties, global demand, and whether Bangladesh exports or imports the crop.

The system also includes smart features such as:

- Crop recommendation based on Bangladesh location (district/region)
- Soil-based crop suggestion system
- Season-based crop selection
- Vitamin-based crop recommendation
- Cooking methods to preserve maximum nutrients
- Country-wise crop profitability and availability analysis
- Crop disease detection with images, symptoms, and solutions

Domain: Agriculture, Education, Nutrition, Information Systems

Real-life context:

In real life, agricultural information is scattered across books, websites, and expert knowledge. Farmers and students often lack a single platform where they can access complete crop intelligence including scientific details, economic demand, and nutritional value. This project solves that problem by creating a centralized and structured agricultural knowledge system.

2. Background and Problem Context

Currently, agricultural learning and decision-making face multiple issues:

- Lack of centralized crop information system
- No proper digital platform for crop scientific and economic data
- Students cannot easily access crop origin, varieties, and history
- No system showing export/import status of crops in Bangladesh
- No integration of nutrition, agriculture, and economic analysis
- Limited knowledge about global demand of crops

Difficulties faced:

- Identifying suitable crops for Bangladesh regions is difficult
- Understanding crop global demand and trade status is complex
- Scientific crop data is scattered and hard to access
- Farmers lack proper digital decision support systems
- Nutritional food knowledge is not integrated with agriculture

Importance of data management:

Using a database system ensures:

- Centralized storage of crop, disease, nutrition, and economic data
- Efficient retrieval of crop information using search and filters
- Relationship between crops, countries, vitamins, and diseases
- Dynamic display of crop details based on user interaction
- Structured and scalable agricultural information system

3. Purpose of the Project

The project aims to:

- Create a centralized agricultural knowledge and decision support system
- Provide detailed crop encyclopedia with images and full information
- Display scientific name, origin, varieties, and history of crops
- Show Bangladesh availability, export/import status, and trade data
- Provide crop recommendation based on region, soil, and season
- Enable vitamin-based crop suggestions for nutrition awareness
- Display crop diseases with images, symptoms, and solutions
- Show country-wise crop demand and profitability analysis
- Provide cooking recipes to preserve maximum nutrients
- Enable consultation system between users and agricultural experts
- Provide profit calculation and decision support tools

Beneficiaries:

- Agricultural students
- Beginner farmers
- Researchers
- Nutrition-focused users
- Agricultural institutions and trainers

4. Learning Outcomes

By completing this project, the following learning outcomes are expected:

- Understanding of relational database design and normalization

- Practical knowledge of SQL queries and database relationships
- Implementation of CRUD operations in a real-world system
- Knowledge of building dynamic web applications using database
- Understanding of data-driven decision support systems
- Ability to design multi-module systems with complex relationships
- Experience in integrating agriculture, nutrition, and economics in one system

5. Proposed Users of the System

Types of Users:

1. Admin

➤ Roles:

- System authentication and login
- Add/Edit/Delete crop encyclopedia data
- Manage crop images and detailed information
- Update scientific data, origin, and history of crops
- Manage disease database with images and solutions
- Control country-wise crop demand and trade data
- Manage vitamin and nutrition database
- Handle user consultations and replies
- Manage system analytics and content updates

2. User / Visitor

➤ Roles:

- View crop encyclopedia with images
- Click on crops to view full detailed information
- Search crops by Bangladesh region, soil, and season
- View crop diseases with images and solutions
- Use vitamin-based crop suggestion system
- View country-wise crop demand and trade information
- Use profit calculator system
- View cooking recipes for maximum vitamin retention
- Send consultation messages and receive expert replies

The system uses role-based access control to separate admin and user functionalities.

6. Initial Idea of Technologies and Tools

Database: MySQL (relational database system, free and widely used)

Backend Language: PHP (Used for authentication, dynamic content handling, and database operations)

Frontend:

- HTML (structure of pages)
- CSS (styling and layout)
- Bootstrap (responsive and professional design components)

Editor: Visual Studio Code

Server: XAMPP (Apache + MySQL + PHP local server environment)

Other Resources:

- Free agricultural and food images from Unsplash and Pexels
- Icons and UI elements from Flaticon

7. Feasibility and Simplicity of the Project

- **Suitable for DBMS course:** The system includes all major DBMS concepts such as relational tables, normalization, joins, and CRUD operations.
- **Completable within academic semester:** Uses free tools and simple technologies, allowing step-by-step development.
- **Beginner-friendly implementation:** Uses standard web technologies without complex frameworks.
- **Highly practical system:** Solves real-world agricultural, economic, and nutritional problems.
- **Scalable design:** Future enhancements like AI prediction and weather integration can be added easily.

8. Conclusion

“**KrishiDisha**” is a comprehensive and advanced database-driven agricultural intelligence platform that integrates crop information, disease detection, nutrition analysis, economic evaluation, and expert consultation into a single system.

The system provides a unique crop encyclopedia feature where users can view detailed agricultural data, including scientific names, origin, Bangladesh availability, export/import status, and global demand. It also supports crop recommendation based on location, soil type, and season, along with vitamin-based nutrition guidance and cooking optimization.

This project demonstrates a strong and practical application of DBMS concepts in a real-world scenario. It is designed to be informative, interactive, and scalable, making it suitable for academic evaluation and future development.